DSC1001 / GE1356 INTRODUCTION TO DATA SCIENCE · OPTION 2: ANALYSE DATA SOLUTIONS · GROUP XX

Who Gets a Recycling Point?

Comparing three collection-point solutions across Hong Kong's 18 districts

[Member 1, ID] · [Member 2, ID] · [Member 3, ID] · [Member 4, ID] · [Member 5, ID] · [Member 6, ID]

22%
of household waste recycled; target ≈55% by 2035

3.2×
gap between districts, mostly from park and office bins

0 / 3
solutions win on every measure

3 of 20
income patterns hold up in every check; most are really size and housing

1 Problem and questions

Hong Kong landfills **10,510 t** of waste a day, and residents say a **nearby bin** is their main reason to recycle (Ma et al., 2023). **How evenly does the network reach each district, and how do the three government solutions compare?**

RQ1 Spread between districts? **RQ2** Solutions compared?
RQ3 Income, and why? **RQ4** Waste or recycling?

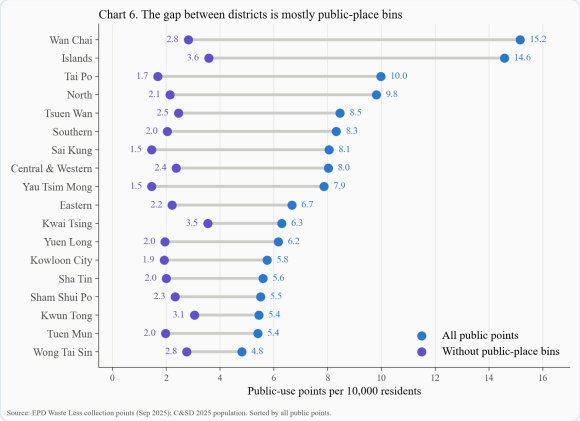
A public-place bins **B** GREEN@COMMUNITY **C** smart bins

2 Data and method

- **8,856** EPD collection points (5,301 public), Sep 2025
- C&SD 2025 population and income; 2021 Census housing; EPD waste statistics 2022–24
- **15** reports and papers converted to common units
- Points **per 10,000 residents**, *n* = 18 districts
- Correlation, "held fixed" checks, six robustness checks incl. a permutation test

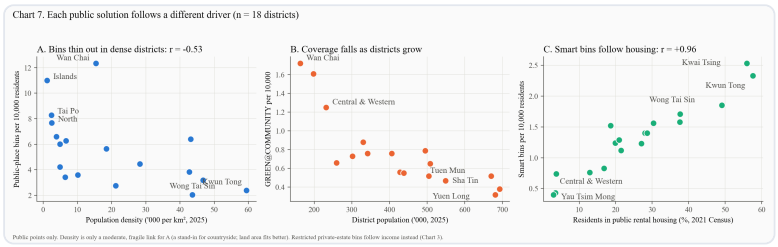
Official aggregated data only; no personal data.

3 What you count changes who looks well served



Wan Chai (15.2) and Islands (14.6) lead only because of bins in parks, offices and hiking roads. Without them, every district falls to 1.5–3.6.

4 Each solution follows a different driver, not income



A thins out in dense districts · **B** falls as districts grow · **C** follows public rental housing (*r* = +0.96). Hold size or housing fixed and the income link nearly vanishes (B: +0.49 → +0.04; C: −0.67 → +0.01).

5 Three solutions, no single winner

Chart 5. Three recycling solutions compared: no single winner

	A. Public-place bins	B. GREEN@COMMUNITY	C. Smart bins
Coverage public points per 10,000 residents	4.79	0.65	1.43
Materials accepted kinds per point	2.6	8.4	1.2
Evenness across districts lower = more even	0.52	0.50	0.43
Opening hours	*Not stated (open-air)*	Stores 10-13 h a day; Spots 3 h a week	80% open 7 am-midnight
Who it reaches	Parks, offices, countryside	Thinnest in the biggest districts	Public rental estates
Tonnes collected a year	*Not published*	47,500	up to 20,000 (food waste)
Cost per tonne	*Not published*	HK$10,100	*Not published*

Darker = better on that measure (rows 1-3). Grey = no published figure. Compare within a row, not across rows.
Source: EPD Waste Less collection points (Sep 2025; public points A = 3,588, B = 491, C = 1,073); C&SD 2025 population; LegCo LCQ2 (2026) and Panel paper (2024). Tonnes and cost are not like-for-like.

6 How far to trust it

- **Exploratory:** only 3 of 20 income tests are reliable
- **RQ4 unanswered:** EPD district waste data are "indicative only"
- Counts ignore capacity and opening hours
- One-day snapshot, 18 districts, no walking distance
- Associations, not causes

7 Recommendations

1. **Measure access by type**, not total count.
2. **Scale GREEN@COMMUNITY to population** (≈59 more Spots).
3. **Target Stores at older private buildings** with no management.
4. **Publish output and cost** for every solution, by district.

Hong Kong does not simply need more recycling points; it needs to know whom each one reaches.

Responsibilities: [Member 1: …] · [Member 2: …] · [Member 3: …] · [Member 4: …] · [Member 5: …] · [Member 6: …]